DIGITAL SAT & EST MATH COURSES

Rabaa Farrag

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EST I MATH • OFFICIAL EXAM • PRACTICE TEST

EST1 August 2021

Official Exam • Full Practice Test

Algebra, functions, data analysis, probability and geometry from the official EST I paper. Work through all 58 questions under timed conditions, then check your answers with the answer key and the step-by-step solutions at the end.

58	MCQ	90 min
QUESTIONS	MULTIPLE CHOICE	TIME LIMIT

NAME _____ DATE _____ SCORE _____ /58

How to get the most from this practice test

- **Read** every question slowly and underline the key words.
- **Draw** a quick sketch for every geometry question before you solve it.
- **Solve** step by step and show your work.
- **Check** your answer by substituting it back before you choose.

RF

YOUR MATH COACH

Eng. Rabaa Farrag

Digital SAT & EST Math expert with **23 years of experience**, helping students master math and reach **700+**.

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Practice Questions

58 QUESTIONS

1 **1.** If $\frac{2x-4}{3}-\frac{x+1}{6}=t+1$ and $t=3$, what is the value of x?

A. 27
B. -1
C. 11
D. $\frac{31}{2}$

2 $Z-Z'=a+bi$

In the equation above, a and b are real numbers and i is the imaginary unit such that $i^2=-1$.

If $Z=3+2i$ and $Z'=4-3i$, what is the value of $(Z-Z')^2$?

(A) $2i$
(B) $-24-10i$
(C) $-26-10i$
(D) $-1+5i$

3 Mathew paid X dollars for a play station that was only 30 dollars less than one third of the original price.

What was the original price in dollars?

(A) $X-30$
(B) $X+60$
(C) $3X+90$
(D) $\frac{1}{3}X-30$

4 A truck contains 15 identical boxes that are either red or blue.

The red box weighs 3 kg and the blue box weighs 2 kg.

If the total weight of the boxes is 36 kgs, what is the difference between the red and blue boxes in the truck?

(A) 6
(B) 9
(C) 1
(D) 3

5 A 100-page album costs twice as much as a 50-page album. The cost of three 100-page albums and two 50-page albums is t. How much does a 50-page album cost?

(A) $8t$
(B) $4t$
(C) $\frac{t}{4}$
(D) $\frac{t}{8}$

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6 An enterprise conducted a study on its products and the results showed that when the unit selling price (P) is raised, the number of units sold (U) went down.

This result is modeled by the equation $10P + 2U = 2500$.

Based on this model, the manager decided to decrease the unit selling price from $50 to 45$.

How many more items did he sell?

(A) 75
(B) 50
(C) 25
(D) 10

7 **7.** $\left[(2x-y)^2-(2x+y)^2\right]^2$

Which of the following is equivalent to the expression above?

A. $16x^4 - y^2$
B. $-64x^4y^4$
C. $-8x^2y^2$
D. $64x^2y^2$

8 **8.** If $\frac{3a}{b} \div c = 7$, what is the value of $\frac{bc}{2a}$?

A. $\frac{3}{14}$
B. $\frac{7}{3}$
C. 21
D. $\frac{6}{7}$

9 Which of the following is equivalent to $f(x) = 2x^2 - 12x + 8$?

(A) $(2x-6)^2 - 28$
(B) $2(x-3)^2 - 10$
(C) $2(x-9)^2 + 5$
(D) $4(x-3)^2 - 5$

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10

D. $4(x-3)^2-5$

10. $f(x)=ax^2+(3-b)x-5$

For the function f defined above, a and b are constants. If $f(1)=2$ and $f(2)=-1$, which of the following is the value of $f(-1)$?

A. -22
B. -16
C. 2
D. 11

11

Sam wants to rent a car. He receives the following offers.

Which offer is the best if Sam wants to travel 100 km?

	Fixed amount to be paid	Amount to be paid for each kilometer driven
Offer A	65$	0.50$
Offer B	60$	0.52$
Offer C	55$	0.54$
Offer D	50$	0.56$

(A) Offer A
(B) Offer B
(C) Offer C
(D) Offer D

12

If m is a real parameter different than zero, what is the number of the real roots of the equation $2x(x^2+4)(mx^2+x-m)=0$?

(A) 1
(B) 2
(C) 3
(D) >3

13

$\sqrt{x^2+3}=x-5$

Which of the following could be a solution for the equation above?

(A) 0
(B) 2.2
(C) 5
(D) None of the above

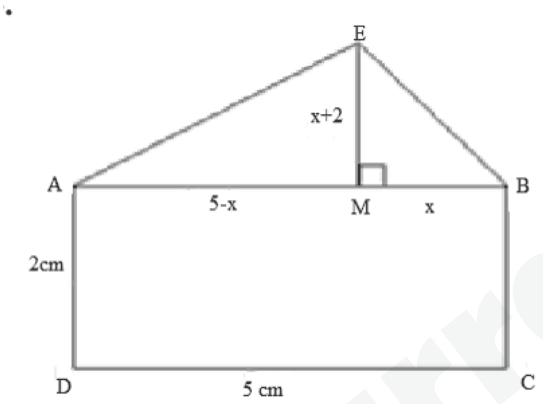
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14 $25x^2 - tx + 4 = (5x - 2)(ax + b)$

In the equation above, a, b and t are constant numbers.
What is the value of t?

(A) 5
(B) -2
(C) 20
(D) -15

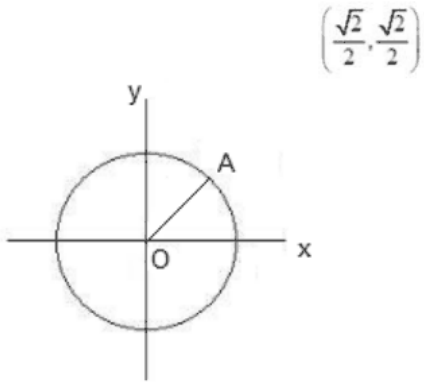
15 Knowing that $x < 5$, use the figure above to find the set of real numbers x if twice the area of triangle BME added to 2 times the area of the triangle MEA is less than or equal to triple the area of rectangle ABCD. (the figure is not drawn to scale)



(A) $x < 4$
(B) $x \leq 4$
(C) $x \in [0, 4]$
(D) $x > 4$

16 If $\frac{3a}{b} \div c = 7$, what is the value of $\frac{bc}{2a}$?

17 In the xy-plane above, O is the center of the circle and the measure of $\angle AOx$ is $\frac{4\pi}{a}$ radians.
What is the value of a?



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18 The table above shows some values for the function f.

If f is a linear function, what is the value of $m + n$?

x	$f(x)$
1	m
2	6
3	n

19 $P(x) = (3a - 6)x^2 + (4 - 2b)x + c - 3$

In the polynomial P above, a, b and c are constant numbers.
If P is identically zero, what is the value of $a + b + c$?

20 $\sqrt{x^2 - 5x + 8} = 2$

What is the product of the two solutions of the equation above?

Section II
Calculator is required
(55 minutes)

6

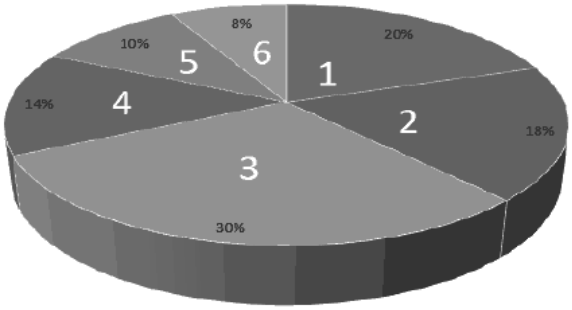
21 In 2017, country Y had 500 miles of paved roads. Starting in 2018, the country has been building 6 miles of new paved roads each year. At this rate, if m is the number of years after 2017, which of the following functions f gives the number of miles of paved roads that will be in country Y assuming that no paved roads go out of service?

A $f(m) = 6 + 2017m$
B $f(m) = 2017 + 6m$
C $f(m) = 500 + 6m$
D $f(m) = 2018 + 6m$

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22 The circular diagram above shows the results of a survey made on the number of books read by 200 pupils in a certain school.

What is the number of pupils who read less than 3 books?



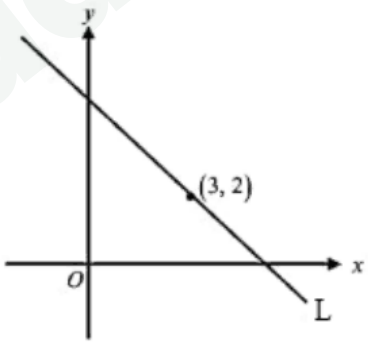
(A) 68
(B) 116
(C) 38
(D) 76

23 A machine finishes the paving of 600m of a road in 1 hour. At the same steady rate, how much time would take two identical machines to finish paving 300m?

(A) 15 minutes
(B) 2 hours
(C) 1 hour
(D) 30 minutes

24 In the xy-plane above, the equation of line L is $2mx - 2y + 12 = 0$, where m is a constant.

What is the slope (gradient) of L?



(A) $-\frac{4}{3}$
(B) $\frac{2}{3}$
(C) $-\frac{8}{3}$
(D) $\frac{1}{2}$

25 Jack has k dollars. He spends $\frac{3}{4}$ of his money on a T-shirt and $\frac{1}{3}$ of what was left on a sandwich. If this left him with t dollars, which of the following is the value of k in terms of t?

(A) $6t$
(B) $9t$
(C) $12t$
(D) $24t$

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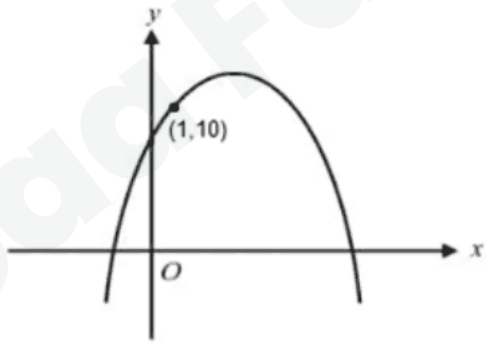
26 The supply function of a product is given by $f(x) = \frac{1+x}{2}$ and the demand function of the same product is given by $g(x) = \frac{2}{x} + 1$, where x represents the price in dollars of the product in both functions. What is the market equilibrium of this product given that it is the point of intersection of the two curves of the two functions?

A) 1.56
B) 1.79
C) 2.56
D) 2.79

27 Tina got on her exams the following grades:
Physics 75/100 with coefficient 2
English 60/100 with coefficient 4
Chemistry 80/100 with coefficient 1
Knowing that the coefficient of Math is 5, what should her minimum grade in Math be to have at least an average of 80/100?

A) 100
B) 85
C) 98
D) She won't be able to have at least this average

28 The graph of $y = tx^2 - 5tx - 6t$ is shown in the xy-plane above, where t is a constant. If the graph passes through the point $(1, 10)$, which of the following is the maximum value of y?



A) 2.5
B) 10
C) 11.75
D) 12.25

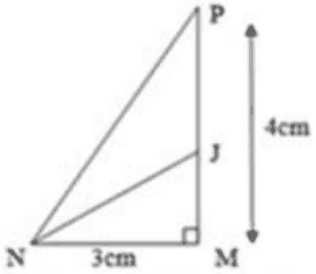
29 A study showed that the number of users of a mobile application has been growing exponentially with the number of new members doubling every 3 months. We know that the initial number of users when the study started was 100,000 users. The equation of the growth is given by $y = 100e^{at}$ $(a \in \mathbb{R})$, measured in thousands of users after t months.

How many users will the application have in 2 years?

A) 25,600 users
B) 25,600,000 users
C) 12,151 users
D) 12,151,041 users

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30 Based on the figure above, what is the area of the triangle MNJ, if the area of the triangle MNJ is double the area of the triangle NJP?



Note: Figure not drawn to scale.

A) 2 cm²
B) 3 cm²
C) 4 cm²
D) 5 cm²

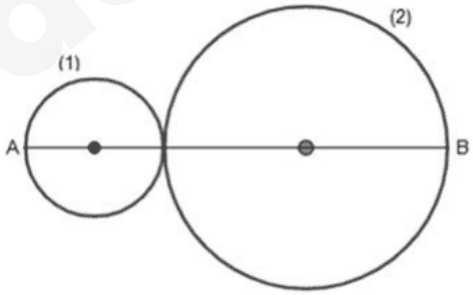
31 $f(x) = (x-2)(x+3) - (2x-4)^2$

Which of the following is an equivalent form of the function above?

A) $(x-2)(11+3x)$
B) $(x-2)(-x+7)$
C) $(x-2)(11-3x)$
D) $(x-2)(x-7)$

32 In the figure above, the circles are tangent to each other. The radius of circle (1) is R and the radius of circle (2) is 2R.

If the sum of their areas is 100π cm², what is the length of [AB]?



A) 100 cm
B) $2\sqrt{5}$ cm
C) 50 cm
D) $12\sqrt{5}$ cm

33 If $3x = 24y$, what is the value of $(\frac{3y}{x})^2$?

A) $\frac{9}{64}$
B) $\frac{3}{4}$
C) $\frac{8}{3}$
D) 24

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34 Tom has horses, dogs and birds in his farm. The number of birds he has is four times the number of dogs, and he has three more dogs than horses.

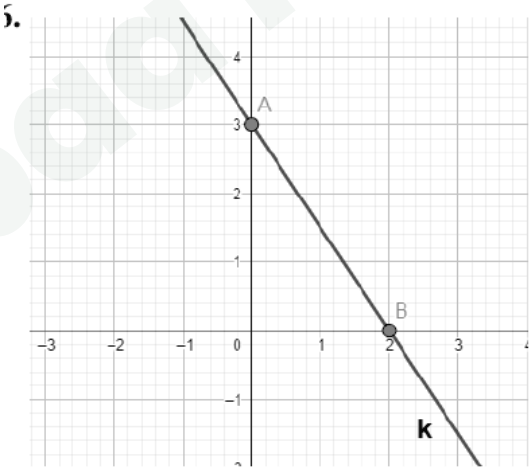
Which of the following could be the total number of these animals?

(A) 25
(B) 26
(C) 27
(D) 28

35 **15.** If x is different than -1, 0 and 1, which of the following is equivalent to $\frac{x}{\frac{1}{x-1}+\frac{1}{1+x}}$?

A. $\frac{1}{x^2-1}$
B. $\frac{(x-1)(x+1)}{2x}$
C. $\frac{2x}{x^2-1}$
D. $\frac{x^2-1}{2}$

36 What is the equation of line d (not shown) that passes through origin O and is perpendicular to line k in the figure above?



(A) $y = \frac{2}{3}x$
(B) $y = -\frac{3}{2}x$
(C) $y = \frac{3}{2}x$
(D) $y = x$

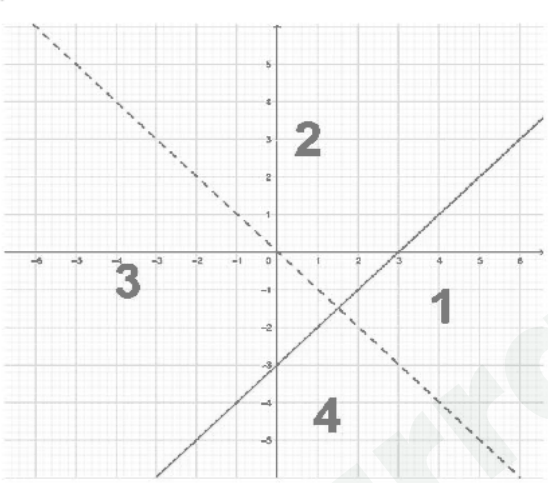
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37 Line m in the xy-plane passes through the points $(2a, a^2)$ and $(2b, b^2)$ with $a \neq b$.

Which of the following is the slope (gradient) of line m?

A) $\frac{a}{b}$
B) $\frac{a+b}{2}$
C) $\frac{2}{a+b}$
D) $\frac{a^2-b^2}{a-b}$

38 What is the solution region of the system $x - y \leq 3$ and $y > -x$ represented in the graph above?



A) Region 1
B) Region 2
C) Region 3
D) Region 4

39 $$\begin{cases} x + 3y = \frac{b}{2} \\ 4x + 4ay = 20 \end{cases}$$

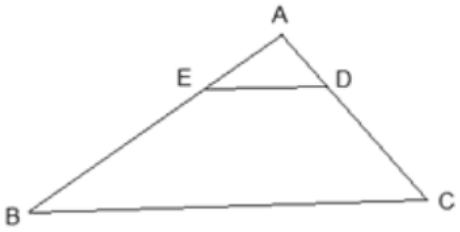
In the system of equations above, a and b are constants. If the system has one solution, which of the following could be the values of a and b?

A) $a = 3; b = 10$
B) $a = 3; b = 12$
C) $a = 3; b = -4$
D) $a = 10; b = 3$

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40 In the figure above, $AE = \frac{1}{3}AB$ and $AD = \frac{1}{3}AC$.

The area of the triangle ABC is how many times the area of the trapezoid EDCB?



Note: Figure not drawn to scale.

(A) $\frac{9}{8}$
(B) $\frac{8}{9}$
(C) 3
(D) $\frac{2}{3}$

41 A container is filled with 200 balls, 80 of them are yellow. After removing x yellow balls, 75% of the remaining balls in the container are not yellow.Which of the following is the value of x?

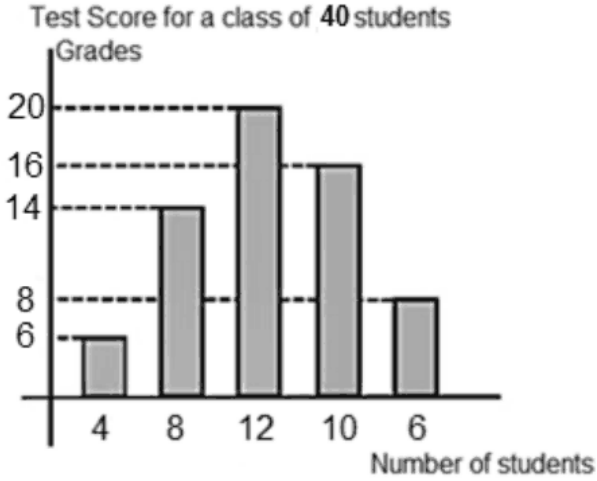
(A) 20
(B) 30
(C) 40
(D) 50

42 In 2007, a watch manufacturer found that 2 out of every 30 watches produced are defected.If the manufacturer produces 2 million watches in a year, which of the following is closest to the estimated number of non-defected watches?

(A) 1,866,000
(B) 1,867,000
(C) 133,000
(D) 134,000

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43 The graph above shows the test grades over 20 of 40 students.Based on the bar graph above, what is the average grade on the test?



(A) 10
(B) 11.6
(C) 12.8
(D) 14.6

44 For what value of x, $|4 - x| - 2$ is less than 0?

(A) -7
(B) 1
(C) 3
(D) There is no such value of x

45 The table above shows the distribution of age and sports activity for 540 employees of a company.

If an employee aged under 40 is selected at random, what is the probability that he practices any kind of sports?

	Practice any kind of sports	No sports activity	Total
Under 40	220	40	260
40 and older	100	180	280
Total	320	220	540

(A) \frac{2}{13}
(B) \frac{27}{11}
(C) \frac{16}{11}
(D) \frac{11}{13}

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46 Ryan estimates that there are x people in a concert.Bella, who knows the actual number of people who attended the concert, y, notes that Ryan's estimate is within 50 people of the actual number of people.Which of the following inequalities represent the relationship between x and y?

	Practice any kind of sports	No sports activity	Total
Under 40	220	40	260
40 and older	100	180	280
Total	320	220	540

A $|x - y| \leq 50$
B $x \leq y + 50$
C $x \geq y - 50$
D $x + y \leq 50$

47

12

Questions 27-28 refer to the following information.

$$f' = \left(\frac{v + v_o}{v}\right) f$$

You are riding in a car at a velocity v_o, in meters per second, towards a loud block party. Because of this movement, the actual frequency of the sound waves emitted by the speakers, f, in hertz, is perceived by you to be a different frequency f', in hertz.

The speaker's sound waves travel at a velocity v, in meters per second. This phenomenon is called the Doppler effect. The formula above shows the relationship between these variables.

27. Which of the following expresses the velocity of the car v_o in terms of the other variables?

A. $v_o = \frac{f' - f}{v f}$
B. $v_o = \frac{f - f'}{f'} v$
C. $v_o = \frac{f}{f'} v$
D. $v_o = \frac{f' - f}{f} v$

48

D. $v_0 = \frac{f' - f}{f} v$

28. If the velocity of the car is 22 m/s, the velocity of the sound waves of the speaker is 340 m/s and you perceive the frequency of the speaker's sound waves to be 500 Hz.

Which of the following is the closest to the actual frequency of the speaker's sound waves?

A. 469
B. 532
C. 470
D. 533

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49 The table below summarizes students' work placement after graduation, based on the major.Given a person who works outside his field of study, which of the following is closest to the probability that he majored in business?

	Work in their field of study	Work in another field	Don't work	Total
Engineering	17,065	10,593	3,867	31,525
Business	18,547	11,753	3,243	33,543
Law	20,372	1,438	542	22,352
Total	55,984	23,784	7,652	87,420

A 0.35
B 0.06
C 0.5
D 0.13

50 $x^2 + y^2 - 6x - 4y + 9 = 0$The equation above is an equation of a circle. Which of the lines represented by the equations below is tangent to this circle?

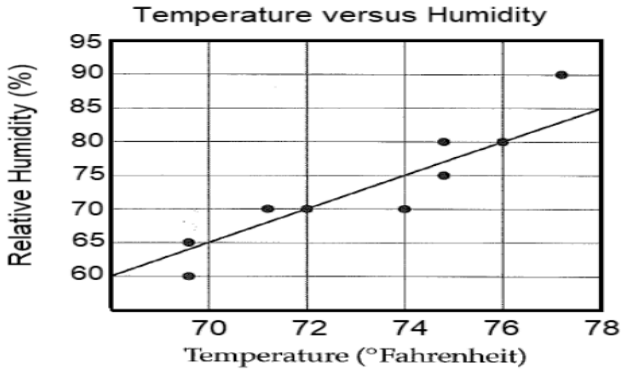
A $y = \frac{1}{3}x - \frac{1}{3}$
B $y = x$
C $y = 3$
D $x = 5$

51 A supermarket has three branches A, B and C in three different cities. The head manager realized that, in average per day, branch A has 20% more customers than branch B and branch B has 20% less customers than branch C.

If the number of customers in branch A is 1200 on a random day, what is the estimated number of customers in branch C that same day?

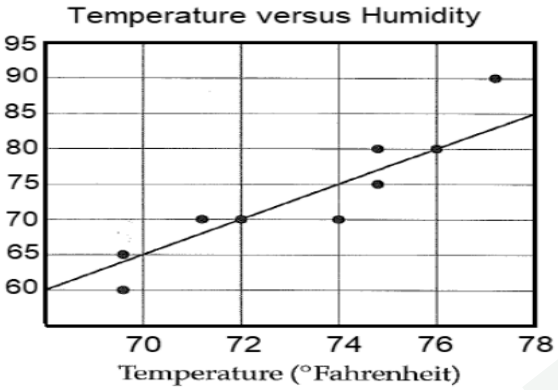
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52



32. Based on the line of best fit, what is the predicted humidity percentage at a temperature of 74° Fahrenheit?

53



33. What is the humidity percentage represented by the data point that is farthest from the line of best fit?

54 James adds to an oil tank 10 liters on day 1, then every day he adds 50% the amount he filled the day before.

After 1000 days, how much oil is there approximately in the tank?

55 Myriam opens a bank account with an initial deposit of 10,000 EGP. The bank account will earn 5 percent interest compounded annually for the first 3 years, after which it will earn 8 percent interest compounded annually.

What is the approximate amount, rounded to the nearest EGP, added to Myriam's account after 5 years?

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56 A spinner has 2 blue sections, 3 red sections and 5 yellow sections. It is spun twice.

What is the probability of getting different colors?

Job title	Commission percentage rate
Trainee	1%
Employee	2%
Supervisor	3%
Senior supervisor	4%
Manager	5%

57

Job title	Commission percentage rate
Trainee	1%
Employee	2%
Supervisor	3%
Senior supervisor	4%
Manager	5%

37. The salary of the manager Sam for the month of January was $7,000. What is the salary of the supervisor John in January?

58

Job title	Commission percentage rate
Trainee	1%
Employee	2%
Supervisor	3%
Senior supervisor	4%
Manager	5%

38. In February, the salary of an employee, a senior supervisor and a trainee was $13,650 altogether.

What was the approximate salary, rounded to the nearest dollar, of the trainee alone in February?

EST I - Mathen

Answer K

NON CALCULATOR

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Answer Key

58 ANSWERS

Q	ANSWER	Q	ANSWER
1	C	30	C
2	B	31	C
3	C	32	D
4	D	33	A
5	D	34	C
6	C	35	D
7	D	36	A
8	A	37	B
9	B	38	B
10	A	39	D
11	D	40	A
12	C	41	C
13	D	42	B
14	C	43	D
15	C	44	C
16	3	45	D
17	16	46	A
18	12	47	D
19	7	48	C
20	4	49	C
21	C	50	D
22	D	51	1250
23	A	52	75
24	A	53	90
25	A	54	20
26	C	55	3503

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Detailed Solutions

58 SOLUTIONS

1 **Answer: C**

See the answer key.

2 **Answer: B** $-24-10i$

$Z-Z'=(3+2i)-(4-3i)=-1+5i$. Then $(-1+5i)^2=1-10i+25i^2=1-10i-25=-24-10i$.

3 **Answer: C** $3X+90$

Let P be the original price. $X=\frac{1}{3}P-30$. Solving for P: $\frac{1}{3}P=X+30\Rightarrow P=3(X+30)=3X+90$.

4 **Answer: D** 3

Let r be the number of red boxes and b be the number of blue boxes. $r+b=15$ and $3r+2b=36$. Multiply first by 2: $2r+2b=30$. Subtract from second: $r=6$. Then $b=9$. The difference is $9-6=3$.

5 **Answer: D** $\frac{t}{8}$

Let x be the cost of a 50-page album. A 100-page album costs $2x$. $3(2x)+2(x)=t\Rightarrow 6x+2x=t\Rightarrow 8x=t\Rightarrow x=\frac{t}{8}$.

6 **Answer: C** 25

At $P=50$, $10(50)+2U=2500\Rightarrow 500+2U=2500\Rightarrow 2U=2000\Rightarrow U=1000$. At $P=45$, $10(45)+2U=2500\Rightarrow 450+2U=2500\Rightarrow 2U=2050\Rightarrow U=1025$. The increase is $1025-1000=25$.

7 **Answer: D**

See the answer key.

8 **Answer: A**

See the answer key.

9 **Answer: B** $2(x-3)^2-10$

$2x^2-12x+8=2(x^2-6x)+8=2(x^2-6x+9-9)+8=2(x-3)^2-18+8=2(x-3)^2-10$.

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10 **Answer: A**

See the answer key.

11 **Answer: D** Offer D

Cost for A: $65 + 0.50(100) = 115$. Cost for B: $60 + 0.52(100) = 112$. Cost for C: $55 + 0.54(100) = 109$. Cost for D: $50 + 0.56(100) = 106$. Offer D is the cheapest.

12 **Answer: C** 3

The roots are $x = 0$, roots of $x^2 + 4 = 0$ (no real roots), and roots of $mx^2 + x - m = 0$. The discriminant of the quadratic is $1^2 - 4(m)(-m) = 1 + 4m^2$. Since $m \neq 0$, $1 + 4m^2 > 0$, so there are 2 distinct real roots for the quadratic. Total real roots = $1 + 2 = 3$.

13 **Answer: D** None of the above

Squaring both sides: $x^2 + 3 = (x - 5)^2 = x^2 - 10x + 25 \Rightarrow 3 = -10x + 25 \Rightarrow 10x = 22 \Rightarrow x = 2.2$. Checking the solution: $\sqrt{2.2^2 + 3} = \sqrt{4.84 + 3} = \sqrt{7.84} = 2.8$. But $x - 5 = 2.2 - 5 = -2.8$. Since $2.8 \neq -2.8$, $x = 2.2$ is an extraneous solution. There is no real solution.

14 **Answer: C** 20

Expanding the right side: $5ax^2 + (5b - 2a)x - 2b$. Comparing coefficients: $5a = 25 \Rightarrow a = 5$. $-2b = 4 \Rightarrow b = -2$. $-t = 5b - 2a = 5(-2) - 2(5) = -10 - 10 = -20 \Rightarrow t = 20$.

15 **Answer: C** $x \in [0, 4]$

Area of BME = $\frac{1}{2}(x)(x + 2)$. Area of MEA = $\frac{1}{2}(5 - x)(x + 2)$. Area of ABCD = $5 \times 2 = 10$. The condition is $2 \times \text{Area}(BME) + 2 \times \text{Area}(MEA) \leq 3 \times \text{Area}(ABCD)$. This simplifies to $x(x + 2) + (5 - x)(x + 2) \leq 30 \Rightarrow (x + 2)(x + 5 - x) \leq 30 \Rightarrow 5(x + 2) \leq 30 \Rightarrow x + 2 \leq 6 \Rightarrow x \leq 4$. Since x represents a length, $x \geq 0$. So $x \in [0, 4]$.

16 **Answer: 3**

$\frac{3a}{bc} = 7 \Rightarrow \frac{a}{bc} = \frac{7}{3} \Rightarrow \frac{bc}{a} = \frac{3}{7}$. Then $\frac{bc}{2a} = \frac{1}{2} \cdot \frac{3}{7} = \frac{3}{14}$.

17 **Answer: 16**

Point A is in Quadrant I with coordinates $(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2})$, which corresponds to an angle of $45°$ or $\frac{\pi}{4}$ radians. We are given the angle measure is $\frac{4\pi}{a}$. So $\frac{4\pi}{a} = \frac{\pi}{4} \Rightarrow a = 16$.

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18 **Answer: 12**

For a linear function, the midpoint of $x = 1$ and $x = 3$ is $x = 2$. Thus, $f(2) = \frac{f(1)+f(3)}{2}$. So $6 = \frac{m+n}{2} \Rightarrow m + n = 12$.

19 **Answer: 7**

If $P(x) = 0$ for all x, then all coefficients are zero. $3a - 6 = 0 \Rightarrow a = 2$. $4 - 2b = 0 \Rightarrow b = 2$. $c - 3 = 0 \Rightarrow c = 3$. Then $a + b + c = 2 + 2 + 3 = 7$.

20 **Answer: 4**

See the answer key.

21 **Answer: C** $f(m) = 500 + 6m$

The initial amount in 2017 is 500 miles. The rate of change is 6 miles per year. So the function is $f(m) = 500 + 6m$.

22 **Answer: D** 76

Less than 3 books means 1 or 2 books. The percentages are 20% and 18%, totaling 38%. 38% of 200 is $0.38 \times 200 = 76$ pupils.

23 **Answer: A** 15 minutes

One machine paves 600m in 60 minutes. Two machines pave $600 \times 2 = 1200$m in 60 minutes. To pave 300m, it takes $300/1200 \times 60 = 15$ minutes.

24 **Answer: A** $-\frac{4}{3}$

The equation is $2y = 2mx + 12 \Rightarrow y = mx + 6$. The line passes through $(3, 2)$. Substituting these coordinates: $2 = m(3) + 6 \Rightarrow 3m = -4 \Rightarrow m = -\frac{4}{3}$. The slope is $m = -\frac{4}{3}$.

25 **Answer: A** $6t$

After spending $\frac{3}{4}$, he has $\frac{1}{4}k$ left. He spends $\frac{1}{3}$ of this on a sandwich, leaving him with $\frac{2}{3}$ of $\frac{1}{4}k$. So $t = \frac{2}{3} \times \frac{1}{4}k = \frac{1}{6}k$. Therefore, $k = 6t$.

26 **Answer: C** 2.56

Set $f(x) = g(x)$: $\frac{1+x}{2} = \frac{2}{x} + 1 \Rightarrow \frac{1+x}{2} = \frac{2+x}{x} \Rightarrow x(1 + x) = 2(2 + x) \Rightarrow x + x^2 = 4 + 2x \Rightarrow x^2 - x - 4 = 0$. Using the quadratic formula: $x = \frac{1 \pm \sqrt{1-4(1)(-4)}}{2} = \frac{1 \pm \sqrt{17}}{2}$. Since price must be positive, $x = \frac{1+\sqrt{17}}{2} \approx \frac{1+4.123}{2} = 2.56$.

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27 **Answer: C** 98

Total coefficient = $2+4+1+5=12$. Total points needed for an 80 average = $80\times 12=960$. Points earned so far = $75(2)+60(4)+80(1)=150+240+80=470$. Points needed in Math = $960-470=490$. Minimum grade in Math = $490/5=98$.

28 **Answer: D** 12.25

Substitute $(1,10)$ into the equation: $10=t(1)^2-5t(1)-6t=t-5t-6t=-10t\Rightarrow t=-1$. The equation is $y=-x^2+5x+6$. The maximum occurs at $x=-\frac{5}{2(-1)}=2.5$. The maximum value is $y=-(2.5)^2+5(2.5)+6=-6.25+12.5+6=12.25$.

29 **Answer: B** 25,600,000 users

The number of users doubles every 3 months. In 2 years (24 months), the number of doubling periods is $24/3=8$. The final number of users is $100,000\times 2^8=100,000\times 256=25,600,000$.

30 **Answer: C** 4 cm²

Triangle NMP is a right triangle at M. NM = 3cm, MP = 4cm. Area of MNJ is double the area of NJP. Since they share the same height (NM), their bases MJ and JP must be in a 2:1 ratio. Thus, $MJ=\frac{2}{3}MP=\frac{2}{3}(4)=\frac{8}{3}$ cm. Area of MNJ = $\frac{1}{2}\times NM\times MJ=\frac{1}{2}\times 3\times\frac{8}{3}=4$ cm².

31 **Answer: C** $(x-2)(11-3x)$

$(x-2)(x+3)-[2(x-2)]^2=(x-2)(x+3)-4(x-2)^2=(x-2)[(x+3)-4(x-2)]=(x-2)[x+3-4x+8]=(x-2)(11-3x)$.

32 **Answer: D** $12\sqrt{5}$ cm

Sum of areas = $\pi R^2+\pi(2R)^2=\pi R^2+4\pi R^2=5\pi R^2=100\pi\Rightarrow R^2=20\Rightarrow R=\sqrt{20}=2\sqrt{5}$. The length of AB is the sum of the diameters: $2R+4R=6R=6(2\sqrt{5})=12\sqrt{5}$ cm.

33 **Answer: A** $\frac{9}{64}$

$3x=24y\Rightarrow x=8y\Rightarrow\frac{y}{x}=\frac{1}{8}$. Then $(\frac{3y}{x})^2=(3\times\frac{1}{8})^2=(\frac{3}{8})^2=\frac{9}{64}$.

34 **Answer: C** 27

Let h be the number of horses. Dogs $d=h+3$. Birds $b=4d=4(h+3)$. Total animals = $h+d+b=h+(h+3)+4(h+3)=2h+3+4h+12=6h+15$. We need $6h+15$ to equal one of the options. If $6h+15=27$, then $6h=12\Rightarrow h=2$. Since h is an integer, 27 is a possible total.

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35 **Answer: D**

See the answer key.

36 **Answer: A** $y = \frac{2}{3}x$

Line k passes through $(0, 3)$ and $(2, 0)$. Its slope is $\frac{0-3}{2-0} = -\frac{3}{2}$. A line perpendicular to k has a slope of $\frac{2}{3}$. Since it passes through the origin $(0, 0)$, its equation is $y = \frac{2}{3}x$.

37 **Answer: B** $\frac{a+b}{2}$

Slope = $\frac{y_2-y_1}{x_2-x_1} = \frac{b^2-a^2}{2b-2a} = \frac{(b-a)(b+a)}{2(b-a)} = \frac{a+b}{2}$.

38 **Answer: B** Region 2

The inequality $x - y \leq 3$ can be rewritten as $y \geq x - 3$, which represents the region above the solid line $y = x - 3$. The inequality $y > -x$ represents the region above the dashed line $y = -x$. The intersection of these two regions (above both lines) is Region 2.

39 **Answer: D** $a = 10; b = 3$

For a system of linear equations to have exactly one solution, the lines must not be parallel, meaning their slopes must be different. The ratio of the coefficients of x and y must not be equal: $\frac{1}{4} \neq \frac{3}{4a} \Rightarrow 4a \neq 12 \Rightarrow a \neq 3$. Only option D has $a \neq 3$.

40 **Answer: A** $\frac{9}{8}$

Triangle ADE is similar to triangle ABC with a ratio of side lengths $\frac{1}{3}$. The ratio of their areas is $(\frac{1}{3})^2 = \frac{1}{9}$. So, Area(ADE) = $\frac{1}{9}$ Area(ABC). The area of trapezoid EDCB is Area(ABC) - Area(ADE) = Area(ABC) - $\frac{1}{9}$ Area(ABC) = $\frac{8}{9}$ Area(ABC). Therefore, Area(ABC) = $\frac{9}{8}$ Area(EDCB).

41 **Answer: C** 40

Total balls = 200, yellow = 80, not yellow = 120.After removing x yellow balls, remaining total = $200 - x$.The number of not yellow balls remains 120.Since 75% of the remaining balls are not yellow:$120 = 0.75(200 - x)$ 120 = 150 - 0.75x \implies 0.75x = 30 \implies x = 40$.

42 **Answer: B** 1,867,000

Defect rate = 2/30 = 1/15. Non-defect rate = 14/15.Number of non-defected watches = $(14/15) \times 2,000,000 = 1,866,666.67$.The closest estimate is 1,867,000.

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43 **Answer: D** 14.6

Since the x-axis sum is 40 (4+8+12+10+6) and matches the total number of students, the x-axis represents the number of students. The y-axis represents the grades (4, 8, 12, 16, 20).Sum of all grades = $(4 \times 8) + (8 \times 14) + (12 \times 20) + (10 \times 16) + (6 \times 6) = 32 + 112 + 240 + 160 + 36 = 580$.Average = $580/40 = 14.5$.The closest given option is 14.6.

44 **Answer: C** 3

$|4 - x| - 2 < 0 \implies |4 - x| < 2 - 2 < 4 - x < 2 \implies -6 < -x < -2 \implies 2 < x < 6$The value 3 is the only option that lies in this interval.

45 **Answer: D** \frac{11}{13}

Out of 260 employees under 40, 220 practice sports. The probability is $220/260 = 11/13$.

46 **Answer: A** $|x - y| \leq 50$

The phrase 'within 50 people' means the absolute difference between the estimate x and actual y is at most 50: $|x - y| \leq 50$.

47 **Answer: D**

See the answer key.

48 **Answer: C**

See the answer key.

49 **Answer: C** 0.5

Working outside their field corresponds to 'Work in another field'. Total people in this category = 23,784.Number of them who majored in Business = 11,753.Probability = 11753 / 23784 \approx 0.494$, which is closest to 0.5.

50 **Answer: D** $x = 5$

Complete the square to find center and radius:$(x^2 - 6x + 9) + (y^2 - 4y + 4) = 4 \implies (x - 3)^2 + (y - 2)^2 = 4$ The circle has center $(3, 2)$ and radius $r = 2$.The distance from center $(3, 2)$ to the vertical line $x = 5$ is $|5 - 3| = 2$, which equals the radius, making it tangent.

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51 **Answer: 1250**

$A = 1.2B \implies 1200 = 1.2B \implies B = 1000.$
$B = 0.8C \implies 1000 = 0.8C \implies C = \frac{1000}{0.8} = 1250.$

52 **Answer: 75**

See the answer key.

53 **Answer: 90**

See the answer key.

54 **Answer: 20**

This forms a geometric series with $a = 10$ and $r = 0.5$.
The sum is $S = \frac{a}{1-r} = \frac{10}{1-0.5} = 20$.
After 1000 days, the sum approaches exactly 20 liters.

55 **Answer: 3503**

Amount after 5 years = $10000 \times (1.05)^3 \times (1.08)^2 \approx 13502.73$ EGP.
The amount added (interest earned) = $13502.73 - 10000 = 3502.73$.
Rounded to the nearest EGP is 3503.

56 **Answer: 0.62**

Total sections = $2 + 3 + 5 = 10$.
Probability of getting same colors = $P(B,B) + P(R,R) + P(Y,Y) = \left(\frac{2}{10}\right)^2 + \left(\frac{3}{10}\right)^2 + \left(\frac{5}{10}\right)^2 = 0.04 + 0.09 + 0.25 = 0.38$.
Probability of getting different colors = $1 - 0.38 = 0.62$.
(Can also be gridded in as 31/50).

57 **Answer: 4520**

See the answer key.

58 **Answer: 2407**

See the answer key.

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